

Termeh Tohidi

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SUMMARY

M.Sc. graduate in Biological Sciences with hands-on experience in **molecular biology, microbiology, sample preparation, experimental design, data analysis, and laboratory documentation**. Skilled in **PCR, DNA extraction, aseptic technique, sample processing**, and maintaining accurate **experimental records**. Experienced working independently and collaboratively in research environments while following **SOPs** and supporting laboratory projects from planning through **data collection and analysis**.

SKILLS

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|-----------------------------------|---|--|
| • DNA Extraction and Purification | • Microbial Culturing and Aseptic Technique | • Biological Sample Handling |
| • PCR Amplification | • Micropipetting | • Technical Documentation |
| • Agarose Gel Electrophoresis | • Centrifugation | • Laboratory Documentation |
| • DNA Sequencing Workflows | • Sample Processing and Preparation | • Laboratory Troubleshooting |
| • Experimental Optimization | | • Data Analysis and Scientific Reporting |

WORK EXPERIENCE

Graduate Research Assistant | *University of Calgary*

May 2021 – Aug 2025

- Conducted M.Sc. thesis research titled “T7-Like Phages Infecting Native Rhizobia: Isolation, Host Range, and Genomic Insights” under the supervision of Dr. Michael Hynes.
- Prepared microbiological media, reagents, and buffers to support bacterial culturing, phage isolation, and molecular biology experiments.
- Collected, labeled, processed, stored, and analyzed biological samples while maintaining accurate laboratory records, experimental documentation, and contamination control procedures.
- Extracted and purified bacterial and phage DNA for **PCR, Nanopore sequencing**, and downstream molecular analyses; assessed DNA quality and concentration using **NanoDrop** and **Qubit** spectrophotometers.
- Optimized experimental conditions for phage isolation, host range analysis, **PCR workflows**, and bacterial culture procedures through systematic testing and evaluation of results.
- Performed **PCR amplification** of **16S rRNA**, **recA**, and **atpD** gene targets and purified PCR products for sequencing and downstream molecular biology applications.
- Performed **restriction enzyme digestion, agarose gel electrophoresis**, and plasmid profiling using **Eckhardt gel electrophoresis** to characterize bacterial and phage genetic material.
- Prepared DNA samples for **Nanopore sequencing** and supported downstream genomic analysis using **SnapGene** and **BLAST**.
- Collected and isolated *Rhizobium* strains from root nodules of seven native plant species using **aseptic microbiological techniques**.
- Isolated 20 bacteriophages from environmental soil samples using **double-layer agar methods** and characterized host range and biological properties.
- Prepared high-titer phage lysates using bench-top, refrigerated, and **ultracentrifugation** techniques.
- Performed routine laboratory procedures including **micropipetting, centrifugation**, sample preparation, reagent preparation, and laboratory equipment operation in support of research activities.
- Collected, recorded, analyzed, interpreted, and reported experimental data while maintaining accurate laboratory documentation and supporting research objectives.
- Followed established **SOPs** and maintained detailed experimental records, laboratory notebooks, and technical documentation.
- Assisted with troubleshooting experimental challenges by identifying potential sources of error, evaluating results, and implementing corrective adjustments to protocols.

- Courses Taught:
 - * CMMB 343 – Microbiology (Winter 2025, Winter 2022)
 - * BIOL 311 – Principles of Genetics (Fall 2023, Spring 2023, Fall 2022)
 - * BIOL 241 – Energy Flow in Biological Systems (Fall 2021)
- Guided students in identifying mutant phenotypes in *Drosophila melanogaster* using dissecting microscopes and analyzing inheritance patterns through simulated **monohybrid** and **dihybrid crosses**
- Instructed students on yeast genetics experiments including haploid strain mating, **complementation testing**, and phenotypic data analysis
- Supervised **bacterial conjugation** experiments and assisted with interpreting gene transfer patterns using selective minimal media
- Taught **DNA extraction**, **PCR**, and **gel electrophoresis**; incorporated **Benchling** for sequence visualization and primer binding site identification
- Trained students in classical microbiology techniques including microscopy, Gram staining, and microbial identification using **selective and differential media**
- Led labs on microbial growth and metabolism under varying environmental conditions, including oxygen availability and nutrient limitation
- Instructed students in antimicrobial sensitivity testing, including the **Kirby-Bauer disk diffusion method** and phenol coefficient determination
- Demonstrated **plaque assays** and one-step growth curve analysis using *E. coli* and **bacteriophage T4**; supported interpretation of viral replication dynamics
- Facilitated **transposon mutagenesis** experiments using plasmid vectors to explore gene function and antibiotic resistance
- Delivered instruction in **genetics**, **microbiology**, and **molecular biology** labs focusing on hands-on training, experimental design, and data interpretation

EDUCATION

University of Calgary

M.Sc. Biological Science

May 2021 - Aug 2025

GPA: 3.5/4.0

Islamic Azad University of Iran

B.Sc. Cellular & Molecular Biology Specialized in Genetics

2012 - 2016

GPA: 3.4/4.0

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Biosafety (Program)
- Laboratory Safety
- WHMIS 2015: *Workplace Hazardous Materials Information System*
- Spill Response
- Occupational Health and Safety Orientation
- Hazard Assessment
- Incident Reporting and Investigation
- R Wizardry Course
- Introduction to Cybersecurity
- Privacy Awareness
- Harassment and Violence Awareness
- **eLearning Foundations of GMP**
 - Quality Risk Management
 - Medicinal Products
 - Active Pharmaceutical Ingredient (API)
 - Deviations, Root Cause Analysis (RCA) Tools and Corrective and Preventive Action (CAPA)
 - Validation

PRESENTATIONS & CONFERENCES

- **Oral Presentation:** “Isolation and Characterization of Rhizobial Phages” – 8th Annual BGSA Symposium
- **Poster Presentation:** Canadian Society of Microbiologists (CSM) 2023, Halifax, NS
- **Poster Presentation:** Canadian Society of Microbiologists (CSM) 2024, London, ON

REFERENCES

Available upon request.